CORE PURE / COMPLEX NUMBERS

Complex Numbers Basics

Mark scheme 13 questions · 94 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

Question	Marks by part	Marks	Page
Question 1	(a)(i) 1 (a)(ii) 2 (a)(iii) 3 (b) 3	9	4
Question 2	(a) 3 (b) 3	6	5
Question 3	(a)(i) 1 (a)(ii) 2 (b)(i) 3 (b)(ii) 1	7	6
Question 4	(a) 2 (b) 4	6	7
Question 5		5	8
Question 6		5	9
Question 7	(a) 1 (b) 5	6	10
Question 8	(a) 2 (b) 3	5	11
Question 9	(a) 2 (b) 2 (c) 3	7	12
Question 10	(a) 2 (b) 6	8	13
Question 11	(a)(i) 2 (a)(ii) 2 (b)(i) 3 (b)(ii) 3	10	14
Question 12	(a) 4 (b) 5 (c) 2	11	15
Question 13	(a) 2 (b) 3 (c) 2 (d) 2	9	16
	Total	**94**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

ft	follow through Correct work from the student's own earlier wrong value earns the mark.	**AG**	answer given The answer is printed in the question, so every step to it must be shown.
cao	correct answer only No follow-through from an earlier error.	**cso**	correct solution only No error anywhere in the part, not just in its answer.
oe	or equivalent	**awrt**	answers which round to
soi	seen or implied Earned if written, or if later working shows it was used.	**isw**	ignore subsequent working A correct answer, once reached, isn't spoiled by later working.
SC	special case Credit, as the note says, for work outside the main method.	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [$\boldsymbol{n}$] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of $\boldsymbol{g}$

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)(i)	$2z - w = 7 - 8i$	**B1**	
		[1]	
1(a)(ii)	$(4-3i)(1+2i) = 4 + 8i - 3i - 6i^2$	**M1**	Expands to four terms and uses $i^2 = -1$. Condone one slip.
	$= 10 + 5i$	**A1**	
		[2]	
1(a)(iii)	$\frac{z}{w} = \frac{(4-3i)(1-2i)}{(1+2i)(1-2i)}$	**M1**	Multiplies numerator and denominator by $1-2i$, or a multiple of it, and attempts to expand.
	$= \frac{-2-11i}{5}$	**A1**	Correct numerator $-2-11i$, or the matching multiple of it for a multiple of $1-2i$.
	$= -\frac{2}{5} - \frac{11}{5}i$	**A1**	oe such as $-0.4 - 2.2i$. $\frac{-2-11i}{5}$ alone scores A0.
		[3]	
1(b)	$p(4-3i) + q(1+2i) = 10 - 13i$ Equates real and imaginary parts: $4p + q = 10, \quad -3p + 2q = -13$	**M1**	Substitutes z and w and equates real and imaginary parts to form two equations in p and q. Condone sign slips.
	$\Rightarrow p = \ldots,\ q = \ldots$	**dM1**	Solves their two linear equations for both p and q, by any method, including a calculator.
	$p = 3, \quad q = -2$	**A1**	
		[3]	
	Total for Question 1	**9**	

Question	Scheme	Marks	Notes
2(a)	$(z-3)^2 - 9 + 34 = 0$ $\Rightarrow (z-3)^2 = \ldots \Rightarrow z = 3 \pm \sqrt{\ldots}$	**M1**	Completes the square, or uses the correct formula with their values, leading to $z = \ldots$. Condone sign slips.
	$(z-3)^2 = -25$	**A1**	Or $\sqrt{-100} = 10i$ oe in the formula.
	$z = 3 + 5i, \quad z = 3 - 5i$	**A1**	Both roots, in the form $a + bi$; $3 \pm 5i$ accepted.
		[3]	
2(b)	$(z^2 + 9)(z^2 - 4) = 0 \Rightarrow z^2 = \ldots$	**M1**	Solves a three-term quadratic in z^2, or in a substituted letter, by any method, leading to $z^2 = \ldots$.
	$z^2 = 4$ or $z^2 = -9$	**A1**	Either correct value of z^2.
	$z = \pm 2, \quad z = \pm 3i$	**A1**	All four roots and no others.
		[3]	
	Total for Question 2	**6**	

Question	Scheme	Marks	Notes
3(a)(i)	$z^* = x - iy$	**B1**	
		[1]	
3(a)(ii)	$zz^* = (x+iy)(x-iy)$ $= x^2 - i^2y^2$	**M1**	Expands $(x+iy)(x-iy)$ and uses $i^2 = -1$. Cross terms may cancel unseen; condone one sign slip.
	$= x^2 + y^2$, which is real	**A1 AG**	Correct expansion leading without error to $x^2 + y^2$, with a statement that it is real.
		[2]	
3(b)(i)	$w = x + iy, \quad w^* = x - iy$, where $x, y \in \mathbb{R}$ $2x + 12 + 2iy = 5x + (14 - 5y)i$	**M1**	Substitutes $w = x + iy$ and $w^* = x - iy$ and equates real and imaginary parts. Alternatively, conjugates the whole equation to obtain $2w^* + 12 = 5w - 14i$, including the sign change of $14i$. Condone one sign slip.
	$2x + 12 = 5x, \quad 2y = 14 - 5y$ $\Rightarrow x = 4, \quad y = 2$	**dM1**	Solves their real and imaginary equations for x and y. For the conjugation method, solves their two equations simultaneously, eliminating w^*, to reach $w = \ldots$. Condone arithmetic slips.
	$w = 4 + 2i$	**A1**	In the form $a + bi$ only; $a = 4$, $b = 2$ alone does not score.
		[3]	
3(b)(ii)	$w^2(w^*)^2 = (12 + 16i)(12 - 16i) = 400$	**B1**	
		[1]	
	Total for Question 3	**7**	

Question	Scheme	Marks	Notes
4(a)	$f(3) = 3^3 - 7 \times 3^2 + 17 \times 3 - 15$	**M1**	Substitutes $z = 3$ into $f(z)$. Division by $(z - 3)$ attempted to a remainder is also accepted.
	$= 27 - 63 + 51 - 15 = 0$ $f(3) = 0$, so $(z - 3)$ is a factor	**A1 AG**	Correct substitution evaluated to 0, or zero remainder from correct division, with a conclusion that $(z - 3)$ is a factor. A0 if the conclusion is missing.
		[2]	
4(b)	$z^3 - 7z^2 + 17z - 15 = (z - 3)\left(z^2 + pz + q\right)$ $\Rightarrow p = \ldots,\ q = \ldots$	**M1**	Divides $f(z)$ by $(z - 3)$, compares coefficients or writes the factor by inspection, to reach a three-term quadratic factor. Condone slips.
	$f(z) = (z - 3)\left(z^2 - 4z + 5\right)$	**A1**	
	"$z^2 - 4z + 5$" $= 0$ $\Rightarrow z = \dfrac{4 \pm \sqrt{16 - 20}}{2} = \ldots$	**M1**	Solves their three-term quadratic factor by formula, completing the square or calculator, leading to $z = \ldots$. Not for solving the cubic directly by calculator.
	$z = 3, \quad z = 2 + i, \quad z = 2 - i$	**A1**	All three roots, the complex ones in the form $a + bi$; $2 \pm i$ accepted.
		[4]	
Total for Question 4		**6**	

Question	Scheme	Marks	Notes
5	$4(a+bi+2i)=(1-2i)(a-bi)$ $\Rightarrow 4a+(4b+8)i=(a-2b)+(-2a-b)i$	**M1**	Substitutes $z=a+bi$ and $z^*=a-bi$ and expands the product using $i^2=-1$. Condone sign slips.
	$4a=a-2b$ $4b+8=-2a-b$	**M1 A1**	**M1** Equates real and imaginary parts of their expansion to form two equations in a and b. **A1** Correct pair, oe.
	$3a+2b=0,\ 2a+5b=-8\Rightarrow a=\ldots,\ b=\ldots$	**dM1**	Solves their two linear equations for a and b by any method; a calculator may be used.
	$z=\frac{16}{11}-\frac{24}{11}i$	**A1**	oe, or awrt $1.45-2.18i$. Must be given as z in the form $a+bi$; a and b stated separately do not score.
	Total for Question 5	**5**	

Question	Scheme	Marks	Notes
6	$(a+bi)^2 = \left(a^2 - b^2\right) + 2abi$ $a^2 - b^2 = 21, \quad 2ab = -20$	**M1 A1**	**M1** Squares $a+bi$ and equates real and imaginary parts with $21-20i$. Condone sign slips. **A1** Both equations, oe.
	$a^2 - \frac{100}{a^2} = 21$ $\Rightarrow a^4 - 21a^2 - 100 = 0 \Rightarrow a^2 = \ldots$	**dM1**	Eliminates one unknown to a three-term quadratic in a^2 or b^2 and solves it by any means, or uses $a^2 + b^2 = \lvert 21 - 20i \rvert$. Condone slips.
	$a^2 = 25 \ \left[a^2 \neq -4\right] \Rightarrow a = \pm 5$	**A1**	$a = \pm 5$, or $b = \pm 2$, with the negative value of a^2 or b^2 rejected or not used.
	$z = 5 - 2i, \quad z = -5 + 2i$	**A1**	Both, and no others; $\pm(5-2i)$ accepted.
	Total for Question 6	**5**	

Question	Scheme	Marks	Notes
7(a)	$2 + 3i$	**B1**	
		[1]	
7(b)	$(z - (2 - 3i))(z - (2 + 3i))$ $= (z - 2)^2 - (3i)^2$	**M1**	Multiplies out the factors from the conjugate pair, or uses their sum and product, to form a quadratic. Condone sign slips.
	$= z^2 - 4z + 13$	**A1**	
	$z^3 + pz^2 + qz + 13$ $=$ ("$z^2 - 4z + 13$")$(z + b)$ $\Rightarrow 13b = 13 \Rightarrow b = \ldots$	**M1**	Writes the cubic as their quadratic times a linear factor and compares a coefficient, or uses a root relationship such as product of roots $= -13$.
	Third root $z = -1$	**A1**	
	$(z^2 - 4z + 13)(z + 1) = z^3 - 3z^2 + 9z + 13$ $p = -3, \quad q = 9$	**A1**	Both values.
		[5]	
Total for Question 7		**6**	

Question	Scheme	Marks	Notes
8(a)	(i) $\lvert z_1 z_2 \rvert = 2\sqrt{3}$	**B1**	
	(ii) $\arg(z_1 z_2) = -\frac{2\pi}{3}$	**B1**	Must lie in $-\pi < \theta \leq \pi$.
		[2]	
8(b)	$\arg(w^n) = n\left(\text{“}-\frac{2\pi}{3}\text{”}\right) = 2k\pi \Rightarrow n = \ldots$	**M1**	Multiplies their $\arg w$ by n and sets it equal to a multiple of 2π, or finds arguments of successive powers until one is. Not for a multiple of π alone.
	(i) $n = 3$	**A1**	
	(ii) $\lvert w^3 \rvert = (\text{“}2\sqrt{3}\text{”})^3 = 24\sqrt{3}$	**B1ft**	ft their $\lvert z_1 z_2 \rvert$ raised to their n.
		[3]	
Total for Question 8		**5**	

Question	Scheme	Marks	Notes
9(a)	$(4+3i)(-12+5i) = -48+20i-36i+15i^2$	**M1**	Expands to four terms and uses $i^2=-1$. Condone one sign slip.
	$-63-16i$	**A1**	
		[2]	
9(b)	$\lvert z_2\rvert = \sqrt{(-12)^2+5^2} = 13$ $\arg z_2 = \pi - \arctan\frac{5}{12}$	**M1**	Attempts the modulus and an argument placed in the second quadrant, such as $\pi - \arctan\frac{5}{12}$. Condone a slip in the modulus.
	$z_2 = 13(\cos\theta + i\sin\theta)$ $\theta = \pi - \arctan\frac{5}{12}$ $[\approx 2.75]$	**A1**	Form $13(\cos\theta+i\sin\theta)$ with θ exact oe or awrt 2.75.
		[2]	
9(c)	$\arg z_1 = \arctan\frac{3}{4}$ $\arg z_2 = \pi - \arctan\frac{5}{12}$ $\arg(z_1z_2) = \pi + \arctan\frac{16}{63}$	**B1**	All three, exact oe or awrt 0.644, 2.75, 3.39, each in $0 \le \arg z < 2\pi$ and its correct quadrant.
	$\tan\left(\arctan\frac{3}{4} - \arctan\frac{5}{12}\right)$ $= \dfrac{\frac{3}{4}-\frac{5}{12}}{1+\frac{3}{4}\times\frac{5}{12}}$	**M1**	Applies the compound angle formula for tan of a sum or difference of their angles, with their tangents substituted. A decimal comparison alone earns neither this nor the A1.
	$= \frac{16}{63}$ $\arg z_1 + \arg z_2 = \pi + \arctan\frac{16}{63}$ $= \arg(z_1z_2)$	**A1 AG**	Exact $\frac{16}{63}$ obtained without error, then a statement that the sum of arguments equals $\arg(z_1z_2)$. Condone no comment on the range of the angle.
		[3]	
	Total for Question 9	**7**	

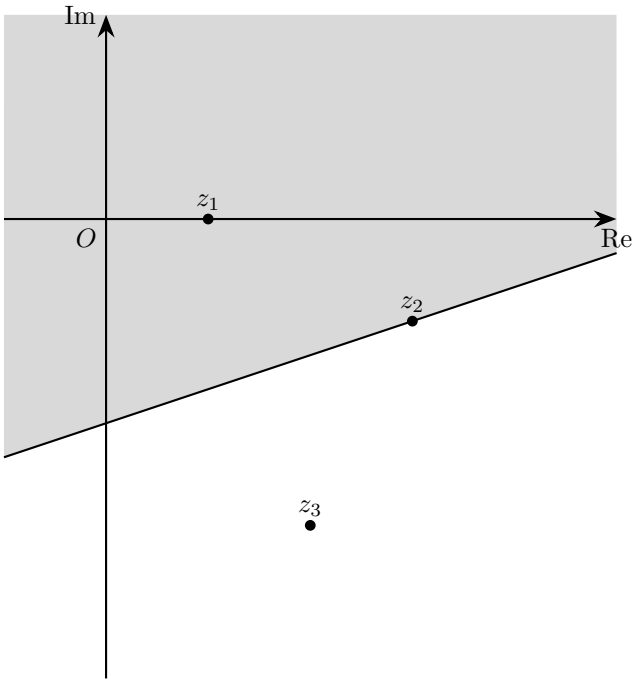
Question	Scheme	Marks	Notes
10(a)	$r = \sqrt{(-3)^2 + 2^2}$ $\theta = \pi - \arctan\frac{2}{3}$	**M1**	Attempts the modulus by Pythagoras and an argument between $\frac{\pi}{2}$ and π using $\arctan\frac{2}{3}$ or $\arctan\frac{3}{2}$.
	$z_1 = \sqrt{13}\,(\cos 2.554 + i \sin 2.554)$	**A1**	$r = \sqrt{13}$ exactly and θ awrt 2.554, in the required form.
		[2]	
10(b)	$z_1 z_2 = (-3 + 2i)(a + bi)$ $= (-3a - 2b) + (2a - 3b)i$	**M1 A1**	**M1** Expands $(-3 + 2i)(a + bi)$ using $i^2 = -1$. Condone one sign slip. **A1** Correct real and imaginary parts, oe.
	$2a - 3b = -(-3a - 2b) \Rightarrow a = \ldots$	**M1**	Uses $\arg = -\frac{\pi}{4}$ to set their imaginary part equal to minus their real part, or their ratio equal to $\tan(-\frac{\pi}{4})$, and finds a relation between a and b.
	$a = -5b$	**A1**	
	$\sqrt{(13b)^2 + (-13b)^2} = 26\sqrt{2} \Rightarrow b = \ldots$	**dM1**	Substitutes their relation into a modulus equation, for $z_1 z_2$ with $26\sqrt{2}$ or for z_2 with $2\sqrt{26}$, and solves for b.
	$z_2 = -10 + 2i$	**A1**	cao. A0 if $10 - 2i$ is also given as an answer.
		[6]	
	Total for Question 10	**8**	

Question	Scheme	Marks	Notes
11(a)(i)	$\lvert w\rvert = \sqrt{(-6\sqrt{3})^2 + (-6)^2}$	**M1**	Takes the square root of the sum of the squares of the real and imaginary parts. Condone sign slips.
	$\lvert w\rvert = 12$	**A1**	
		[2]	
11(a)(ii)	$\arg w = -\pi + \arctan\left(\frac{6}{6\sqrt{3}}\right)$	**M1**	Uses arctan of a ratio of the parts' magnitudes and adjusts for the third quadrant, e.g. $-\pi + \alpha$ or $-\frac{\pi}{2} - \beta$.
	$\arg w = -\frac{5\pi}{6}$	**A1**	$-\frac{5\pi}{6}$ or awrt -2.62, in the range $-\pi < \arg w \leq \pi$.
		[2]	
11(b)(i)	$z_1 z_2 = 2(a\cos\theta + 3\sin\theta) + 2(a\sin\theta - 3\cos\theta)i$ $a\cos\theta + 3\sin\theta = -3\sqrt{3}$ $a\sin\theta - 3\cos\theta = -3$	**M1**	Expands $z_1 z_2$ and equates real and imaginary parts with w, giving two equations; condone sign slips. Or uses $\lvert z_1\rvert = \frac{\lvert w\rvert}{\lvert z_2\rvert}$.
	$(a\cos\theta + 3\sin\theta)^2 + (a\sin\theta - 3\cos\theta)^2$ $= 27 + 9 \Rightarrow a^2 + 9 = 36 \Rightarrow a = \ldots$	**dM1**	Squares and adds their equations to eliminate θ, or sets $a^2 + 9$ equal to their $\lvert z_1\rvert^2$, leading to $a = \ldots$
	$a = 3\sqrt{3}$	**A1**	$3\sqrt{3}$ or $\sqrt{27}$. A0 if the negative root is also given as an answer.
		[3]	
11(b)(ii)	$z_2 = \frac{-6\sqrt{3}-6i}{\text{“}3\sqrt{3}\text{”}-3i} \times \frac{\text{“}3\sqrt{3}\text{”}+3i}{\text{“}3\sqrt{3}\text{”}+3i}$	**M1**	Divides w by z_1 with their a, multiplying numerator and denominator by the conjugate of the denominator. Condone sign slips.
	$z_2 = -1 - \sqrt{3}\,i$	**A1**	Or $\cos\theta = -\frac{1}{2}$ and $\sin\theta = -\frac{\sqrt{3}}{2}$.
	$\theta = -\pi + \arctan\sqrt{3} = -\frac{2\pi}{3}$	**A1**	In the range $-\pi < \theta \leq \pi$.
		[3]	
	Total for Question 11	**10**	

Question	Scheme	Marks	Notes
12(a)	$zw = (3+i)(2+4i)$ $= 6 + 12i + 2i + 4i^2$	**M1**	Expands the product using $i^2 = -1$. Condone one arithmetic slip.
	$zw = 2 + 14i$	**A1**	
	$\lvert zw\rvert = \sqrt{2^2 + 14^2} = \ldots$ $\arg(zw) = \arctan\frac{14}{2} = \ldots$	**M1**	Finds the modulus and argument of their zw, the argument in the correct quadrant for their zw.
	$zw = 10\sqrt{2}\,(\cos(\arctan 7)$ $+ i\sin(\arctan 7))$	**A1**	Accept $\sqrt{200}$ or awrt 14.1, and θ awrt 1.43.
		[4]	
12(b)	$\frac{1}{v} = \frac{z-w}{zw} = \frac{(3+i)-(2+4i)}{\text{“}2+14i\text{”}}$	**M1**	Combines over a common denominator, or finds $\frac{1}{w}$ and $\frac{1}{z}$ in the form $a + bi$ and subtracts.
	$\frac{1}{v} = \frac{1-3i}{2+14i}$	**A1**	oe, such as $-\frac{2+i}{10}$.
	$v = \frac{(2+14i)(1+3i)}{(1-3i)(1+3i)}$	**M1**	Inverts their $\frac{1}{v}$ and multiplies numerator and denominator by the conjugate of the denominator.
	$= \frac{2+6i+14i+42i^2}{1-9i^2}$	**dM1**	Expands numerator and denominator using $i^2 = -1$, reaching a real denominator. Condone one slip.
	$v = -4 + 2i$	**A1**	
		[5]	
12(c)	$vv^* = (-4+2i)(-4-2i) = \ldots$	**M1**	Multiplies their v by its conjugate, or uses $vv^* = a^2 + b^2$ with their a and b.
	$vv^* = 20$, so $k = 20$	**A1**	
		[2]	
	Total for Question 12	**11**	

Question	Scheme	Marks	Notes
13(a)	Real coefficients $\Rightarrow 3+i$ and $2+3i$ also roots	**E1**	States that non-real roots occur in conjugate pairs, so the conjugates of z_2 and z_3 must also be roots.
	Five distinct roots, but a quartic has at most four	**E1**	Concludes that five distinct roots, or more than four, is impossible for a degree-four polynomial.
		[2]	
13(b)	$\frac{2-i}{1-3i} \times \frac{1+3i}{1+3i} = \ldots$	**M1**	Substitutes and multiplies numerator and denominator by the conjugate of their denominator. Condone sign slips in forming the differences.
	$= \frac{1}{2} + \frac{1}{2}i$	**A1**	oe, such as $\frac{5+5i}{10}$.
	First quadrant $\Rightarrow \arg = \arctan 1 = \frac{\pi}{4}$	**A1 AG**	Correct quotient leading without error to $\frac{\pi}{4}$, with the first quadrant, or both parts positive, stated.
		[3]	
13(c)	$\arg\left(\frac{z_2-z_1}{z_3-z_1}\right) = \arg(2-i) - \arg(1-3i)$ $= -\arctan\frac{1}{2} - (-\arctan 3)$	**M1**	Uses the argument of a quotient as the difference of arguments, with arctan forms of both arguments placed in the fourth quadrant. Condone one sign slip.
	$\frac{\pi}{4} = \arctan 3 - \arctan\frac{1}{2}$	**A1 AG**	Correct negative arguments equated to $\frac{\pi}{4}$ from (b), reaching the printed result without error.
		[2]	

Question 13 continued

Question	Scheme	Marks	Notes
13(d)			
	Shape	**B1**	Straight line through the midpoint of z_1 and z_3, visibly perpendicular to their join. Intercepts need not be marked.
	Shading	**B1**	Half-plane containing z_1 shaded, on a line that is at least roughly the bisector. Condone a dashed boundary.
		[2]	
	Total for Question 13	**9**	